

1. Model Definition

Relays form a Cox process supported in the vacant set of a fixed radius Boolean model in $\mathbb{R}^2$.

They are connected when they can “see” each other. We call the resulting network the Visibility-Relay Graph: $\mathcal{G}(X^\lambda; \mu, r_0)$.

We say that the model *percolates* if there exists an **infinite connected component** in the **Visibility-Relay Graph** (VRG).

Notation: λ is the intensity of the **relays**, μ is the intensity of the **obstacles** and r_0 is the **radius** of the obstacles.

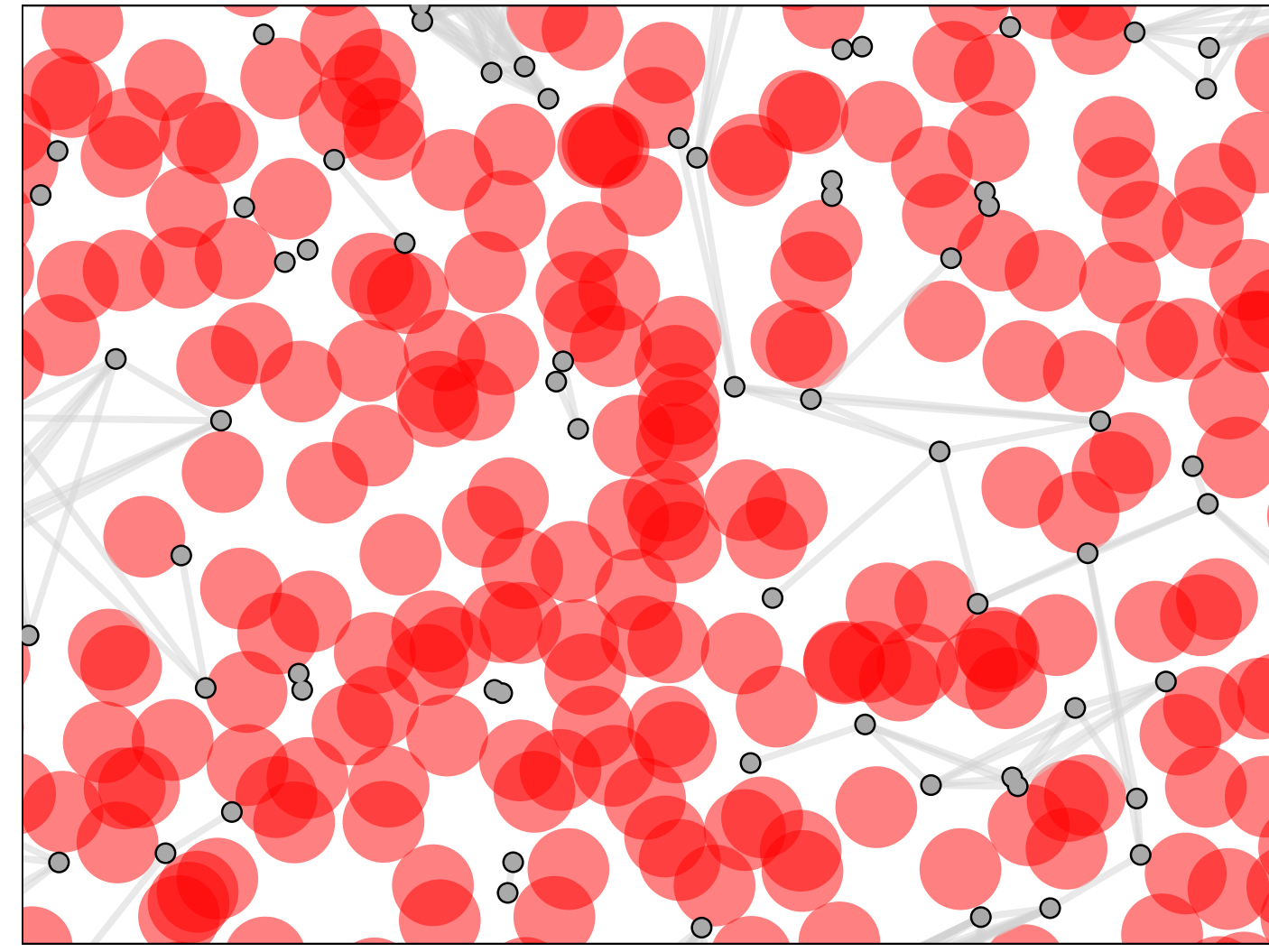


Figure 1: A realization of the model

2. Main Result

The model is monotone in the relay intensity. It is therefore natural to define the critical relay intensity

$$\lambda_c(\mu, r_0) := \inf \{ \lambda > 0 : \mathbb{P}(\mathcal{G}(X^\lambda; \mu, r_0) \text{ percolates}) > 0 \}.$$

Main Theorem (Non trivial phase transition)

For every $r_0 > 0$ and $0 < \mu < \mu_c(r_0)$, the Visibility-Relay Graph has a non-trivial phase transition:

$$0 < \lambda_c(\mu, r_0) < +\infty$$

3. Proof Strategy

Both bounds on λ_c follow from **discretization**, **renormalization**, **finite-range dependence** and comparison with **independent site percolation**.

Subcritical regime $\lambda_c(\mu, r_0) > 0$

Discretization (subcritical case)

For the subcritical regime, the discretization chosen is a hexagonal lattice. $H_n(z)$ is the flat hexagonal cell centered on $z \in \mathbb{T}$ of sidelength n . For $y \in \mathbb{R}$, we define R_y the maximal visibility of this point.

Definition : n -bad cells

$H_n(nz)$ is n -bad if:

1. $R(H_n(nz)) := \sup_{y \in H_n(nz)} R_y < n/2$;
2. $H_n(nz) \cap X^\lambda = \emptyset$.

A cell which is not n -bad is called n -good.

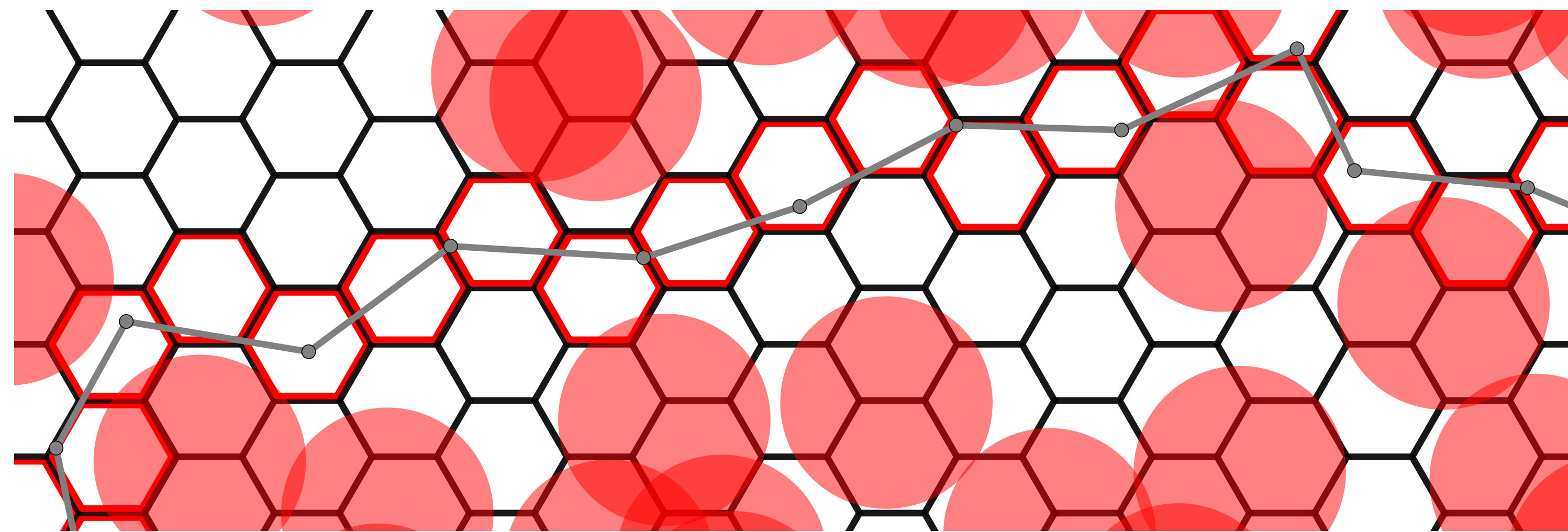


Figure 2: The path in the VRG implies a path of n -good cells

This definition of n -goodness is chosen such that:

- ▶ Path in the Visibility-Relay Graph $\implies$ n -good path in the hexagonal grid
- ▶ $\mathbb{P}(H_n(z) \text{ is } n\text{-good}) \xrightarrow[n \rightarrow \infty, \lambda \rightarrow 0^+]{}$ 0

k -dependence

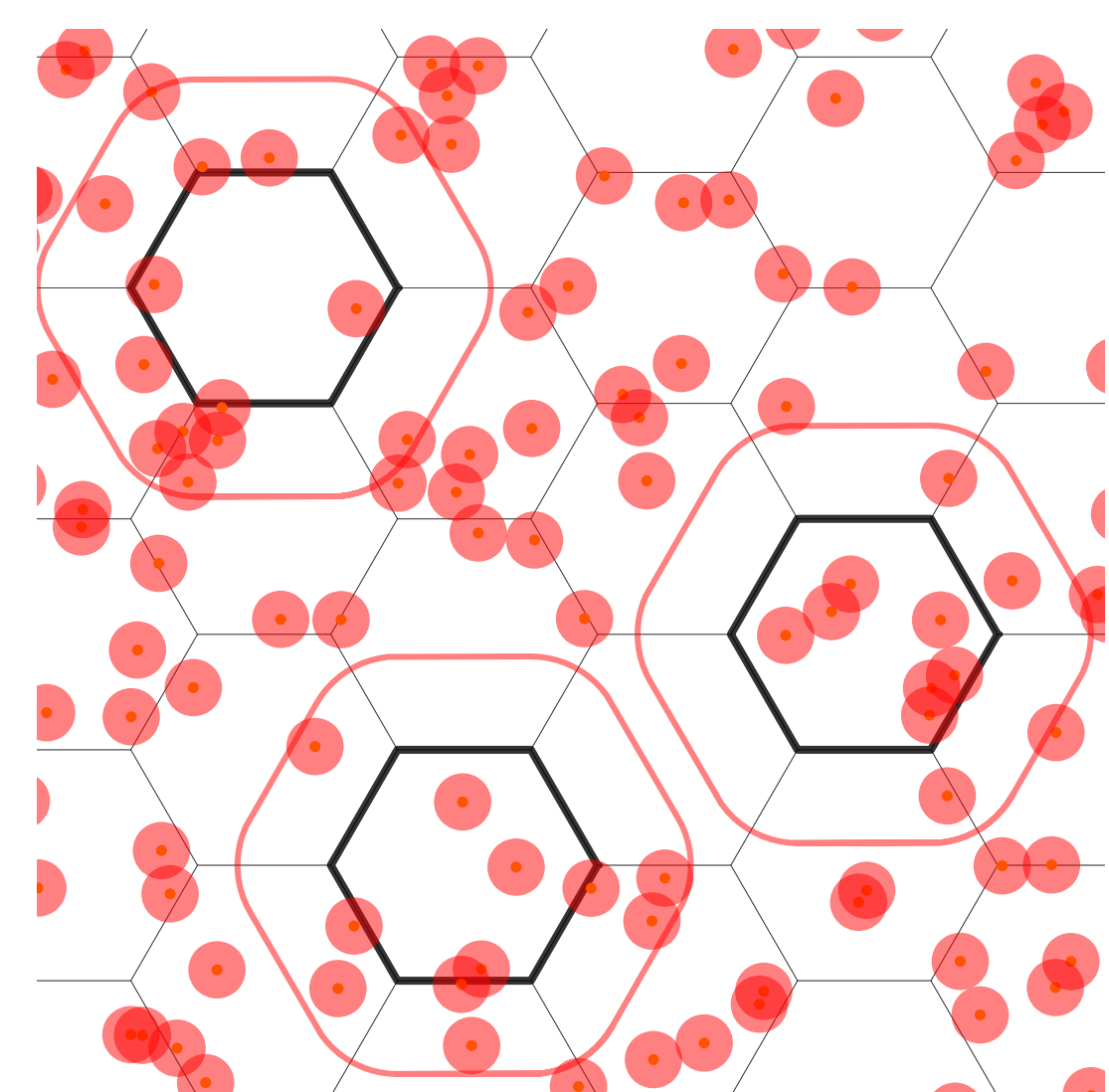


Figure 3: Hexagonal lattice

In both cases, for cells at graph distance at least 3, the n -goodness events are independent.

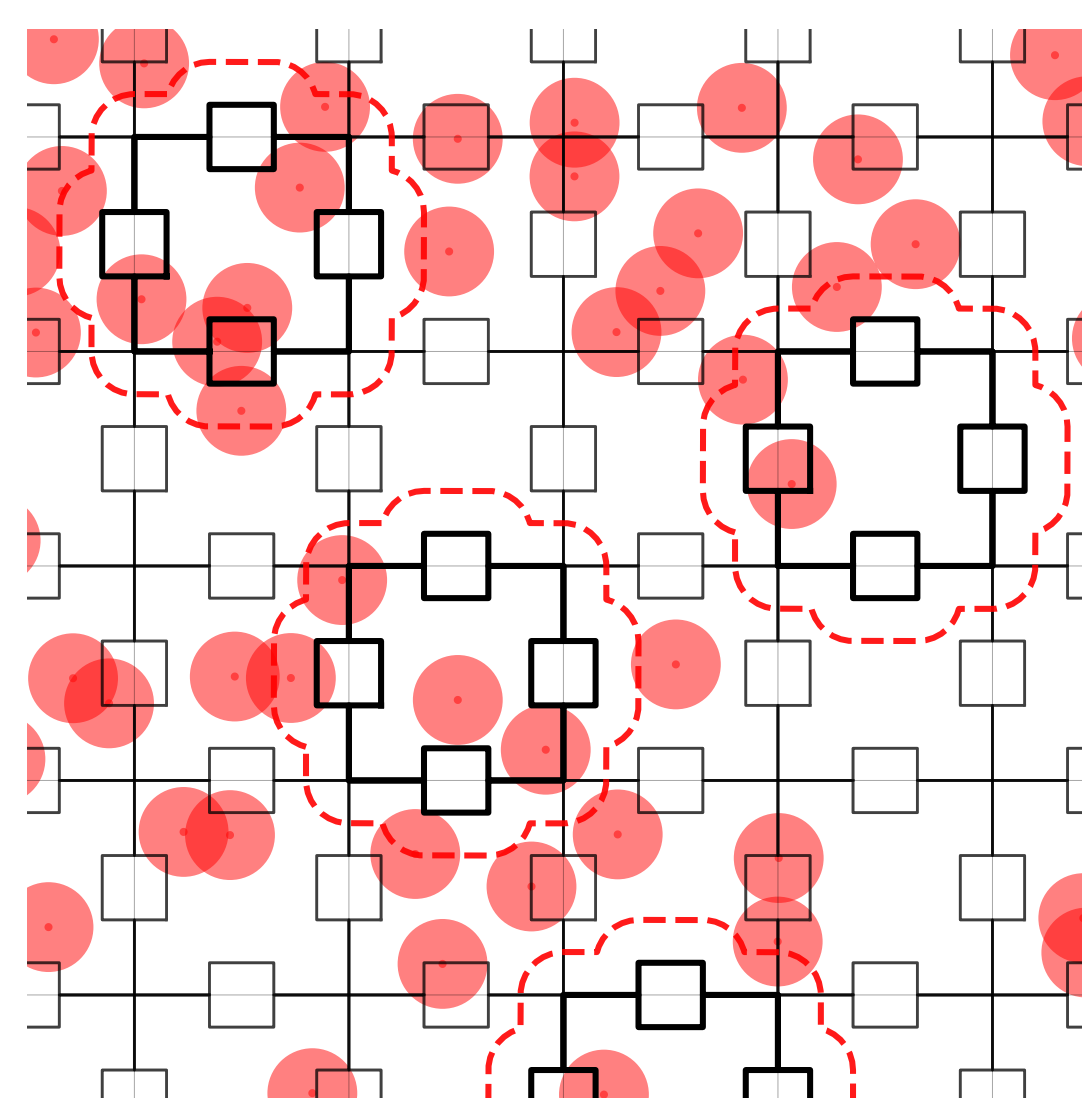


Figure 4: Square lattice

Supercritical regime $\lambda_c(\mu, r_0) < +\infty$

Discretization (supercritical case)

For the supercritical regime, the discretization chosen is a square lattice. $Q_n(z)$ is the square cell centered on $z \in \mathbb{Z}^2$ of sidelength n .

Definition : n -good cells

For $z \in \mathbb{Z}^2$, the cell $Q_n(nz)$ is called n -good if:

1. It contains the required vacant crossings in $\mathcal{V}_{r_1}$ for $r_1 > r_0$
2. Every small box intersecting the cell and inside $\mathcal{V}_{r_0}$ contains at least one relay (the size of the small boxes depends on $r_1 - r_0$)

A cell which is not n -good is called n -bad.

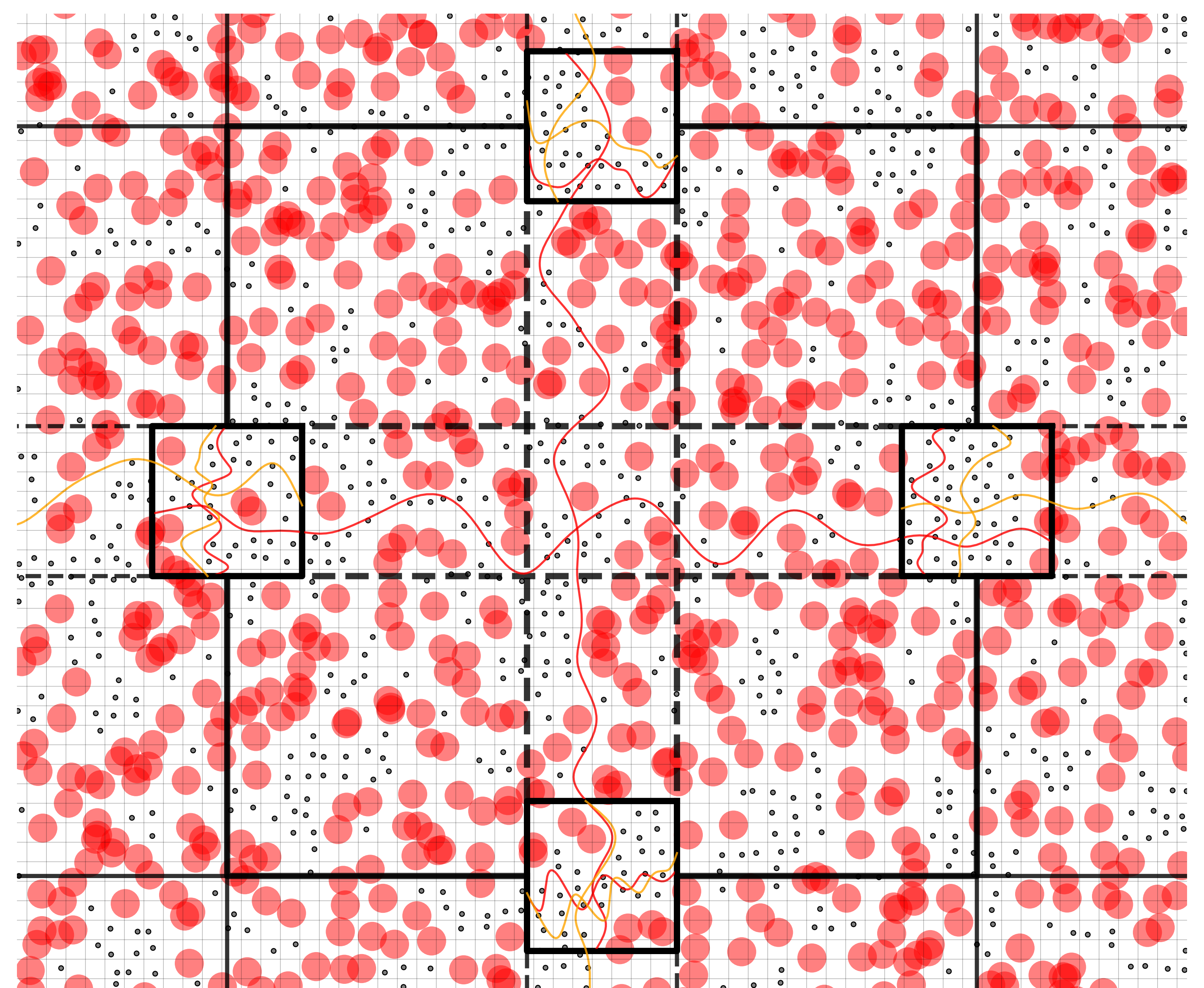


Figure 5: An n -good cell

The definition of n -goodness is such that

- ▶ Path of n -good cells in the square grid $\implies$ Path in the Visibility-Relay Graph.
- ▶ $\mathbb{P}(Q_n(nz) \text{ is } n\text{-good}) \xrightarrow[n, \lambda \rightarrow +\infty]{}$ 1

Final argument

$$\mathbb{P}(n\text{-good}) \xrightarrow[n \rightarrow +\infty]{\lambda \rightarrow 0} 0 \text{ and } k\text{-dependence}$$

LSS domination theorem

The n -good process does not percolate and
VRG path $\implies$ n -good path

No percolation in the VRG

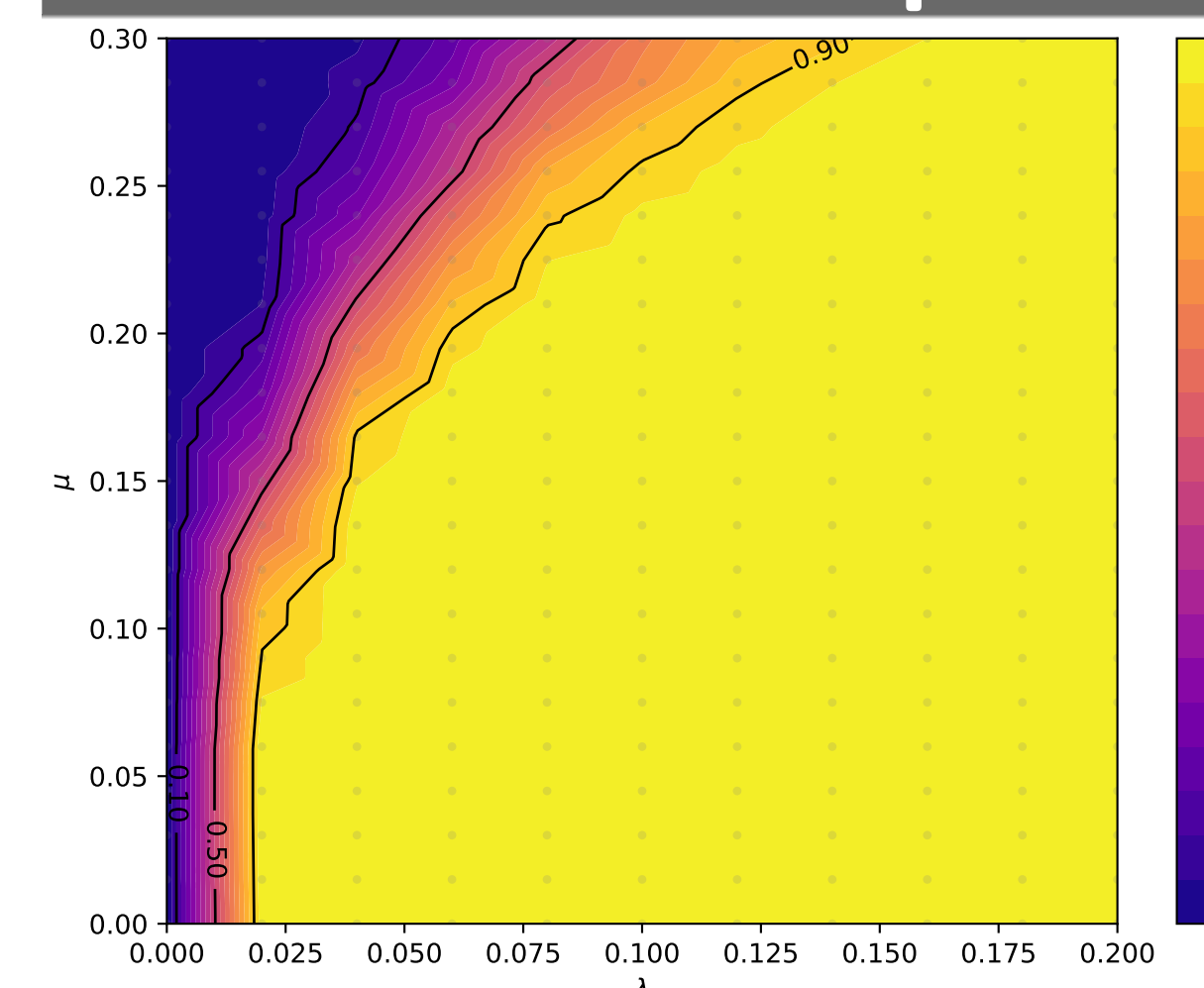
$$\mathbb{P}(n\text{-good}) \xrightarrow[n \rightarrow +\infty]{\lambda \rightarrow +\infty} 1 \text{ and } k\text{-dependence}$$

LSS domination theorem

The n -good process percolates and
 n -good path $\implies$ VRG path

Percolation in the VRG

4. Numerical Exploration



Finite-size simulations: We numerically estimate finite-sized window crossing probabilities.
Open Question: This method works for estimating the critical density of the vacant set but we have not proven yet that this is an estimator for $\lambda_c(\mu, r_0)$. This is related to the *sharpness* of the transition, something we want to try to prove next.

References

- [1] Hirsch, Jahnke, Cali (2017). “Continuum Percolation for Cox processes”
- [2] Calka, Michel, Porret-Blanc (2009). “Visibilité dans le modèle Booléen”
- [3] Liggett, Schonmann, Stacey (1997). “Domination by Product Measure”

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